

Recap

We defined

- Tautology
- Contradiction
- Contingency

Examples

$p \vee \neg p$ is Tautology

$p \wedge \neg p$ is a Contradiction

p is a Contingency

Definition

Two propositions, S and T are logically equivalent, iff $S \leftrightarrow T$ is a Tautology.

in this case we write $S \equiv T$

The symbol $\equiv$ for logical equivalence behaves much like an $=$ in arithmetic,

- We can substitute equivalent subexpression for each other
- We can chain together strings of equivalences

Analogy

If $x = 7y$, then $f(x) + x^2 = f(7y) + (7y)^2$

Also if $x = 7y = 3z + 2 = 4r - 3$, then $x = 4r - 3$ etc.

1.3 Logical Equivalence

p	q	$\neg p$	$\neg q$	$p \vee q$	$\neg p \wedge \neg q$	$\neg(p \vee q)$	$(\neg p \wedge \neg q) \leftrightarrow \neg(p \vee q)$
T	T	F	F	T	F	F	T
T	F	F	T	T	F	F	T
F	T	T	F	T	F	F	T
F	F	T	T	F	T	T	T

Since the final column is all T it is a Tautology, thus $\neg p \wedge \neg q \equiv \neg(p \vee q)$

Alternatively, omit the final column and highlight the fact that the two columns are identical

Similiarly $\neg p \vee \neg q \equiv \neg(p \wedge q)$

Collectively $\neg p \vee \neg q \equiv \neg(p \wedge q)$

&

$\neg p \wedge \neg q \equiv \neg(p \vee q)$

Are De Morgan's Law(s) (for propositional logic)

Also

$p \rightarrow q \equiv \neg p \vee q$

Called the conditonal-disjunction equivalence

Tables

Table 6 lists many equivalences involving $\wedge, \vee, \neg$

Table 7 adds identities involving $\rightarrow$

Table 8 includes $\leftrightarrow$

We don't always need to build a truth table at all, we can use properties of $\equiv$ to build new equivalences from old,

Note

In arithmetic we have 2 distributive laws.

$$x * (y + z) = (x * y) + (x * z) \text{ (This is the same pattern as our logic law)}$$

But Also

$$(x + y) * z = (x * z) + (y * z)$$

Example

Can we show

$$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

Approach 1 : build a truth table

Approach 2 : Use existing equivalences,

- Exploration (start at both sides, work to the middle)

$$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (\neg p \vee r) \wedge (\neg q \vee r)$$

$$(p \vee q) \rightarrow r \equiv \neg(p \vee q) \vee r \equiv (\neg p \wedge \neg q) \vee r$$

Approach 3 : This new identity is in table 7, if we hadn't been so busy learning table 6

Explanation Stage

Tidy this up into a linear proof,

Common Logical Equivalences

Here, T means true and F means false.

Law	Logical equivalence
Identity	$p \wedge T \equiv p$ $p \vee F \equiv p$
Domination	$p \vee T \equiv T$ $p \wedge F \equiv F$
Idempotent	$p \wedge p \equiv p$ $p \vee p \equiv p$
Double negation	$\neg(\neg p) \equiv p$
Negation	$p \vee \neg p \equiv T$ $p \wedge \neg p \equiv F$
Commutative	$p \wedge q \equiv q \wedge p$ $p \vee q \equiv q \vee p$
Associative	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ $(p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributive	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
Absorption	$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$
De Morgan's	$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$
Implication	$(p \rightarrow q) \equiv (\neg p \vee q)$
Contrapositive	$(p \rightarrow q) \equiv (\neg q \rightarrow \neg p)$
Biconditional	$(p \leftrightarrow q) \equiv ((p \rightarrow q) \wedge (q \rightarrow p))$

TABLE 6 Logical Equivalences.	
<i>Equivalence</i>	<i>Name</i>
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws